A five-face incidence identity for right-angled hyperbolic polytopes and positive-curvature facet examples

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Abstract

For an ideal right-angled hyperbolic four-polytope Q, put $\kappa(Q) = 3f_0(Q) - 4f_3(Q) + 24$. Every ideal right-angled hyperbolic five-polytope satisfies $\sum_Q \kappa(Q) = 48$, with the sum over its facets. Our internally proved second-volume and equality theorems imply that at least one such facet has volume at least $3\pi^2$. We give analogous necessary incidence inequalities for ideal six-polytopes and mixed polytopes. The stronger proposed assertion that every ideal right-angled four-polytope has nonpositive κ is false: three further facet doublings of the double 24-cell give an exact example with $\kappa = 6$. Iterating doubling makes positive κ unbounded. No new dimension exclusion or five-dimensional example follows. The four-dimensional volume and equality inputs are explicit; these results are not presented as externally refereed or fully formalized.

1 A universal carrier identity

Let Q be an entirely ideal right-angled hyperbolic four-polytope. Define

$$\Delta(Q) = f_0(Q) - f_3(Q), \qquad \kappa(Q) = 3\Delta(Q) - (f_3(Q) - 24).$$

The published volume formula is $\mathrm{Vol}(Q) = (\Delta(Q) + 4)\pi^2/3$ [1]. The ideal regular 24-cell has $(\Delta, \kappa) = (0, 0)$; its reflected facet double has $(\Delta, \kappa) = (4, -2)$. We use the internal second-volume and equality theorems: every other ideal right-angled four-polytope has integer $\Delta \geq 5$. These inputs are proved in the separate manuscripts *The second volume of ideal right-angled hyperbolic four-polytopes* and *The equality structure of the second-volume ideal right-angled hyperbolic four-polytope*; they are not consequences of the identity below.

Theorem 1. *Every entirely ideal right-angled hyperbolic five-polytope P satisfies*

$$\sum_{Q \text{ a facet of } P} \kappa(Q) = 48.$$

Consequently it has a facet Q with $\kappa(Q) > 0$, $\Delta(Q) \geq 5$ and $\mathrm{Vol}(Q) \geq 3\pi^2$.

Proof. Write $(v, e, r, g, f) = (f_0, f_1, f_2, f_3, f_4)(P)$. A cusp link is a rectangular four-box, with sixteen vertices, twenty-four two-faces and eight facets. Counting the corresponding incidences gives

$$e = 8v, \qquad \sum_{R \in \Omega_3(P)} f_0(R) = 24v, \qquad \sum_{Q \in \Omega_4(P)} f_0(Q) = 8v.$$

Each ideal right-angled three-face satisfies $f_2(R) = f_0(R) + 2$. A two-face belongs to three three-faces: its finite relative-interior normal diagram consists of three orthogonal walls. Hence $3r = 24v+2g$. The boundary is a four-sphere, so $v-e+r-g+f = 2$. Eliminating e, r gives $g = 3v+3f-6$.

Each three-face belongs to two four-facets. Therefore

$$D := \sum_Q \Delta(Q) = 8v - 2g = 2v - 6f + 12, \qquad A := \sum_Q (f_3(Q) - 24) = 2g - 24f = 3D - 48.$$

The desired identity is $\sum_Q \kappa(Q) = 3D - A = 48$. A positive summand exists. Neither the 24-cell nor its unique second-volume double has positive κ, so the asserted volume lower bound follows from the separate classification. □

A positive carrier satisfies $f_3(Q) \le 3\Delta(Q) + 23$ and $f_0(Q) \le 4\Delta(Q) + 23$. In the first possible branch $\Delta = 5$, these bounds are respectively 38 and 43; this is a necessary domain, not an existence assertion.

Corollary 2. *For an entirely ideal right-angled hyperbolic six-polytope,*

$$\sum_{Q \in \Omega_4(P)} \kappa(Q) = 24 f_5(P).$$

At least $\lceil f_5(P)/2 \rceil$ *four-faces have positive* κ.

Proof. Apply the theorem to every five-facet. Each four-face has exactly two five-cofaces, giving the identity. Each five-facet contains a positive carrier, and any one four-face can serve only its two cofaces. □

2 Mixed finite-volume polytopes

Let P be any finite-volume right-angled n-polytope, $n \ge 5$, with v_{fin} finite vertices. Write I_5 for its entirely ideal five-faces. A nonideal five-face contains a finite vertex, and a finite vertex is simple, so belongs to $\binom{n}{5}$ five-faces. Thus

$$f_5 - I_5 \le \binom{n}{5} v_{\mathrm{fin}}.$$

A four-face has an orthogonal relative-interior normal diagram of rank $n-4$ and exactly $n-4$ five-cofaces. Summing the carrier identity over ideal five-faces, discarding nonpositive terms and using this multiplicity, we obtain

$$\boxed{48 \left(f_5 - \binom{n}{5} v_{\mathrm{fin}} \right) \le (n-4) \sum_{Q \text{ an ideal four-face}} \max(\kappa(Q), 0).}$$

If h_4 counts positive carriers, the unweighted version is $f_5 \le \binom{n}{5} v_{\mathrm{fin}} + (n-4) h_4$. For $n = 9, 10, 11, 12$, the respective coefficient pairs are $(126, 5), (252, 6), (462, 7), (792, 8)$. No arithmetic restriction is imposed. These inequalities alone do not exclude any of the open dimensions.

3 A positive carrier by actual facet doubling

Use the Lorentz metric $\mathrm{diag}(-1,1,1,1,1)$ and outward unit wall normals. For a facet normal n of an ideal right-angled polytope P, let s_n be its wall reflection. Then $P \cup s_n P$ is exactly the intersection of the halfspaces with normals $m, s_n m$, $m \neq n$. Indeed, for $x \in P$, acuteness gives

$$\langle x, s_n m\rangle = \langle x, m\rangle - 2\langle m, n\rangle\langle x, n\rangle \leq 0.$$

Conversely, a point of the indicated intersection is in P if its product with n is nonpositive, and otherwise reflects into P. The union is therefore convex, with disjoint interiors and doubled volume. Its Klein vertex set is the union of the two ideal vertex sets; every distinct point on the strictly convex sphere remains extreme.

If the gluing facet has b vertices, its $b+2$ neighboring facets merge with their reflected copies, while other facets occur twice. Indeed, if $m = s_n\ell$ are two old normals, acuteness and $\langle m, n\rangle = -\langle \ell, n\rangle$ force both products to be zero, so $m = \ell$. Thus there are no other cross-copy identifications, and

$$v' = 2v - b, \qquad f' = 2f - b - 4, \qquad \kappa' = 2\kappa + b - 8. \tag{1}$$

An adjacent facet with a vertices, meeting the gluing facet in a k-gon, becomes a facet with $2a - k$ vertices.

Start with the reflected double of the regular 24-cell. In the exact rational realization, take the three nine-vertex facets

$$F = (1,0,1,-1,0), \quad G = (1,0,1,0,-1), \quad H = (1,0,1,1,0).$$

Their consecutive products are zero, their consecutive ridges are quadrilaterals, and $\langle F, H\rangle = -1$. Double successively across F, G, H. After doubling F, the G facet has fourteen vertices while H still has nine. After doubling G, the H facet has fourteen. Equation (1) and the complete exact reflection certificate give

24-cell copies	gluing size	v	f	κ
2	–	42	38	-2
4	9	75	63	-3
8	14	136	108	0
16	14	258	198	6.

The final f-vector is $(258, 1032, 972, 198)$ and its volume is $64\pi^2/3$. This refutes $\kappa \leq 0$ universally, without constructing a five-polytope. Since every subsequent gluing facet has at least six vertices, further doubling gives $\kappa_{t+1} \geq 2\kappa_t - 2$ and hence $\kappa_t \geq 4\cdot 2^t + 2$. Positive carrier values are unbounded.

4 Scope and reproduction

The five-face identity and its corollaries have two independent internal mathematical reviews. The positive-carrier construction has exact generation, fresh replay and an independent saved checker; two independent full mathematical reviews have passed. The precise review scopes are recorded in the accompanying repository report. These are necessary incidence constraints and actual four-dimensional examples, not new right-angled dimension bounds. The exact Python scripts use only the standard library. The identity certificate checks linear combinations of universal geometric premises; it does not prove those premises by sampling. The doubling checker imports no producer and verifies each saved reflected normal/vertex set, facet rank, strict center, cube cusp

link and count. Completeness of the original 24-cell vertex census is inherited from the frozen base certificate; the written halfspace-union proof transports completeness through each double. See `docs/right_angled_five_face_carriers_reproduce.md` for commands and the precise internal/external lemma map.

References

[1] A. Kolpakov, *On the optimality of the ideal right-angled 24-cell*, Algebraic & Geometric Topology 12 (2012), 1941–1960. doi:10.2140/agt.2012.12.1941.